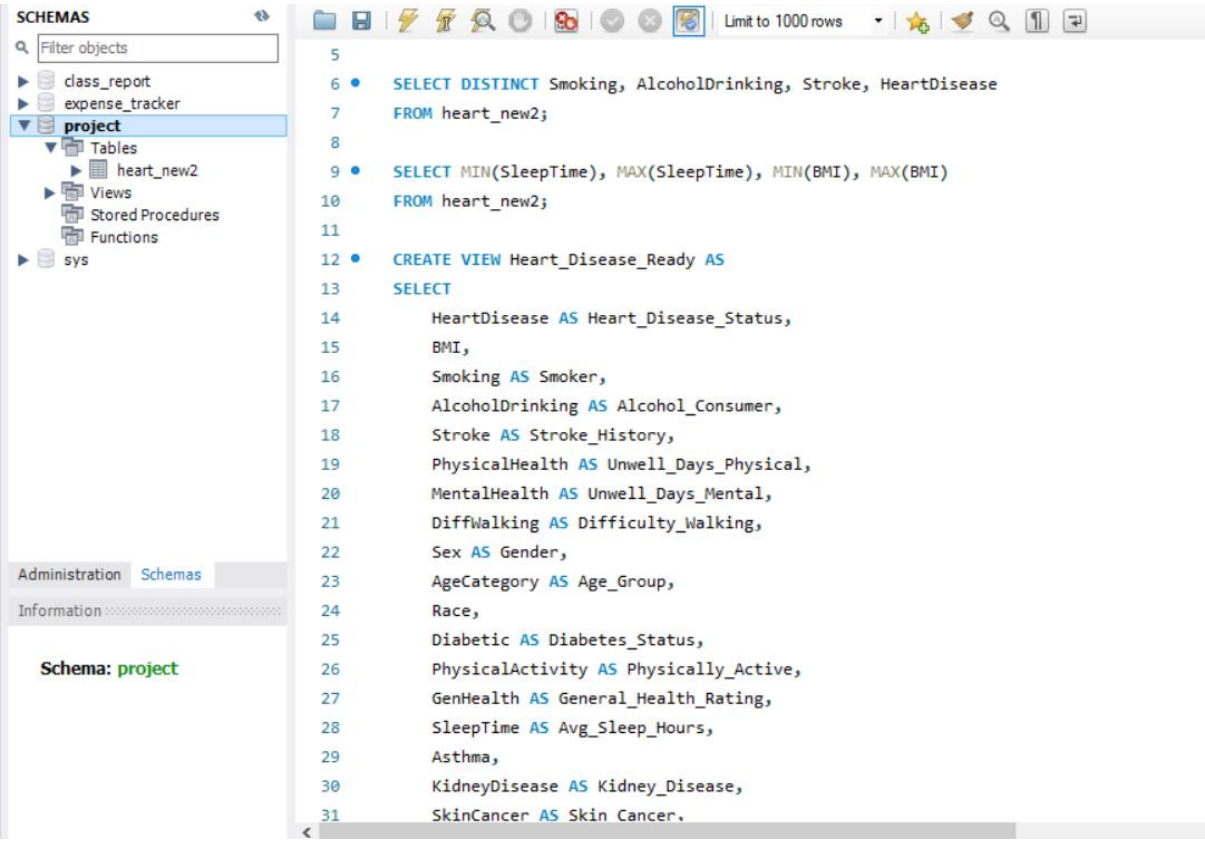
PROJECT DEVELOPMENT PHASE

Date	02 March 2026
Team ID	
Project Name	Heart Disease Analysis
Maximum Marks	8 Marks

Data Connectivity & Preparation

The project utilizes a multi-stage data pipeline starting from a raw dataset to a processed SQL view for Tableau consumption.

- **Dataset Source:** The heart disease dataset containing patient records was acquired for analysis.
- **Database Integration:** The raw data was imported into a MySQL database to ensure relational integrity and structured querying.



- **Data Preparation Steps:**
  - **Renaming & Aliasing:** Technical column names were converted to clinical aliases, such as changing "Diabetic" status to include "Gestational Diabetes" for better clarity.

```sql
use project;
select * from heart_new2;
select HeartDisease, Diabetic from heart_new2;
Select * from heart_new2 order by AgeCategory;

SELECT DISTINCT Smoking, AlcoholDrinking, Stroke, HeartDisease
FROM heart_new2;

SELECT MIN(SleepTime), MAX(SleepTime), MIN(BMI), MAX(BMI)
FROM heart_new2;
```

- **SQL View Creation:** A centralized view was developed using CASE statements to categorize continuous data into logical groups.
- **Preprocessing Logic:** BMI values were grouped into standard clinical categories (Underweight, Healthy, Overweight, Obese) to facilitate demographic risk analysis.

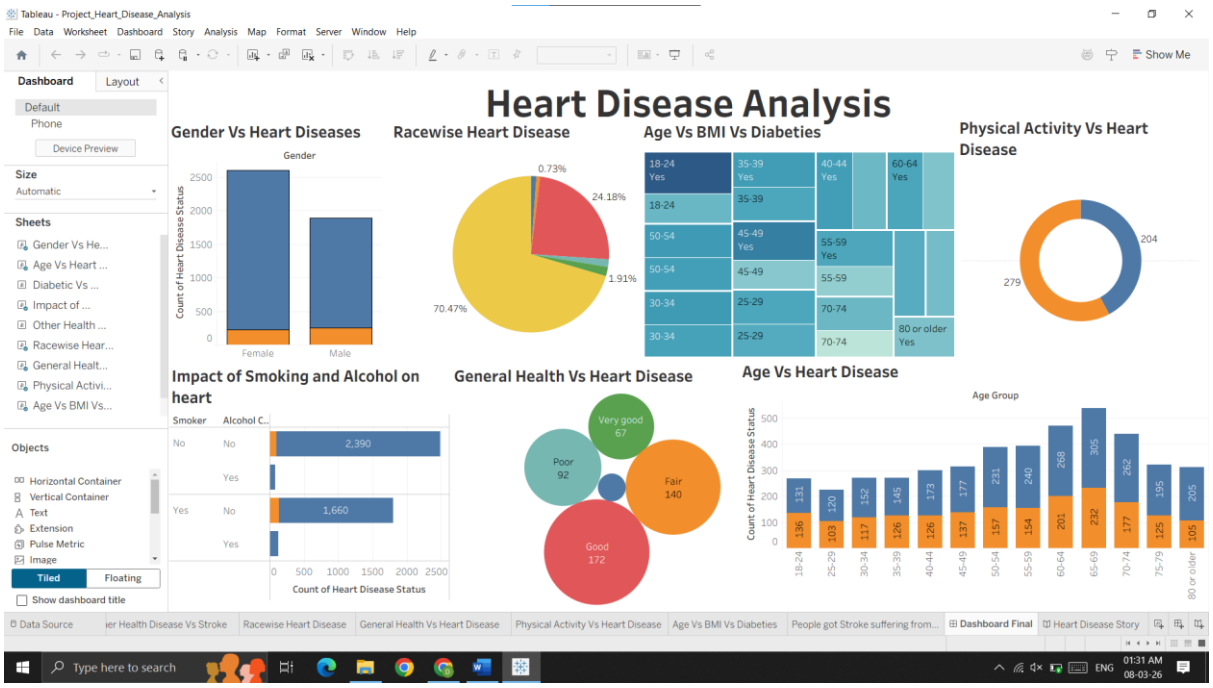
```sql
    -- Logic for BMI Categories
    CASE
        WHEN BMI < 18.5 THEN 'Underweight'
        WHEN BMI >= 18.5 AND BMI <= 24.9 THEN 'Healthy Weight'
        WHEN BMI >= 25.0 AND BMI <= 29.9 THEN 'Overweight'
        ELSE 'Obese'
    END AS BMI_Category,

    -- Logic for Lifestyle Risks
    CASE
        WHEN Smoking = 'Yes' AND AlcoholDrinking = 'Yes' THEN 'High Risk (Both)'
        WHEN Smoking = 'Yes' OR AlcoholDrinking = 'Yes' THEN 'Moderate Risk'
        ELSE 'Low Risk (Neither)'
    END AS Lifestyle_Risk_Profile

FROM heart_new2;

#viewing view
select * from Heart_Disease_Ready;
```

Dashboard & Story Development

The final output consists of a 7-8 visualization dashboard designed for healthcare providers.

- **Calculated Field Logic:** A custom **"Patient Risk Score"** was created by aggregating multiple binary risk factors (Smoking, Alcohol, Stroke, Diabetes) into a 0–5 scale.

- **Visualization Highlights:**
  - **Core Trend:** An area chart visualizing "Age vs. Heart Disease" to establish the primary risk baseline.
  - **Lifestyle Impact:** A combined analysis of smoking and alcohol consumption to show compounding cardiovascular risks.
  - **Clinical Correlation:** A scatter plot mapping BMI and Age against Diabetic status to identify medical "tipping points".



- **Tableau Storytelling:** The workbook is organized into a Tableau Story that guides the user from demographic overviews to specific lifestyle interventions and final clinical risk assessments.

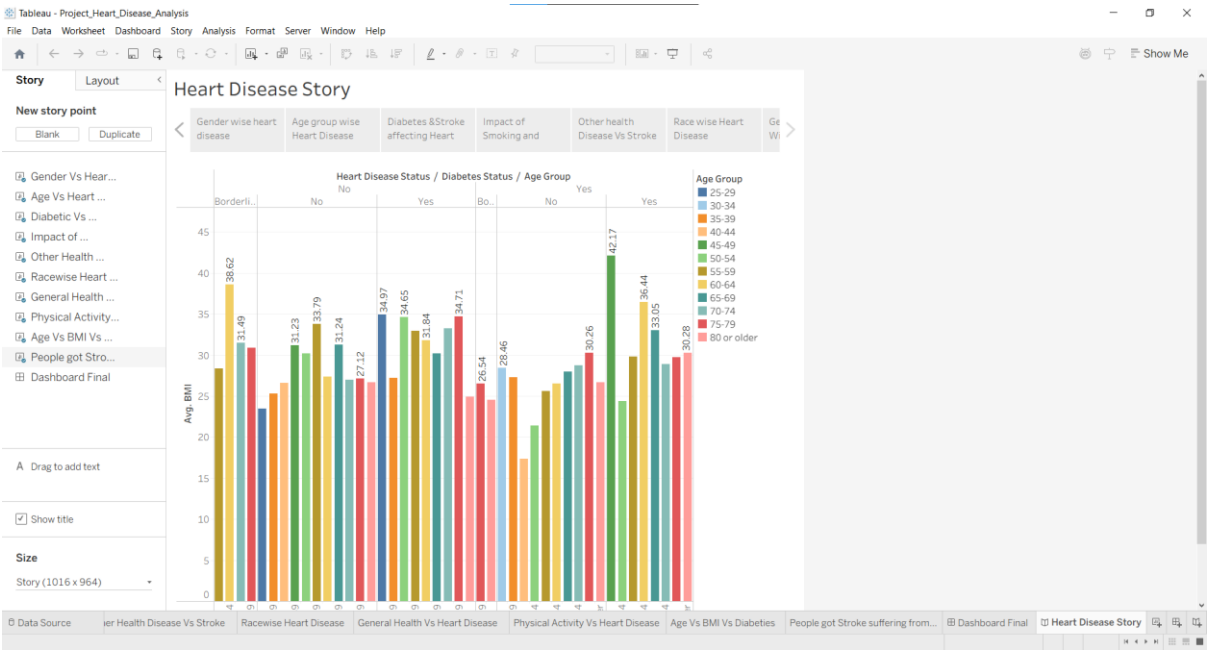


Tableau - Project_Heart_Disease_Analysis
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Show Me
Story
Layout
New story point
Blank
Duplicate
Gender Vs Hear...
Age Vs Heart ...
Diabetic Vs ...
Impact of ...
Other Health ...
Racewise Heart ...
General Health ...
Physical Activity...
Age Vs BMI Vs ...
People got Stro...
Dashboard Final
Drag to add text
Show title
Size
Story (1016 x 964)
Heart Disease Story
Gender wise heart disease
Age group wise Heart Disease
Diabetes &Stroke affecting Heart
Impact of Smoking and
Other health Disease Vs Stroke
Race wise Heart Disease
Heart Disease Status / Diabetes Status / Age Group
No
Yes
Borderli..
Bo..
Avg. BMI
Age Group
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80 or older
38.62
31.49
31.23
33.79
31.24
27.12
34.97
34.65
31.84
34.71
26.54
28.46
30.26
42.17
36.44
33.05
30.28
Data Source
Racewise Heart Disease
General Health Vs Heart Disease
Physical Activity Vs Heart Disease
Age Vs BMI Vs Diabeties
People got Stroke suffering from...
Dashboard Final
Heart Disease Story